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DATA ANALYTICS WITH AI INTERNSHIP 2026

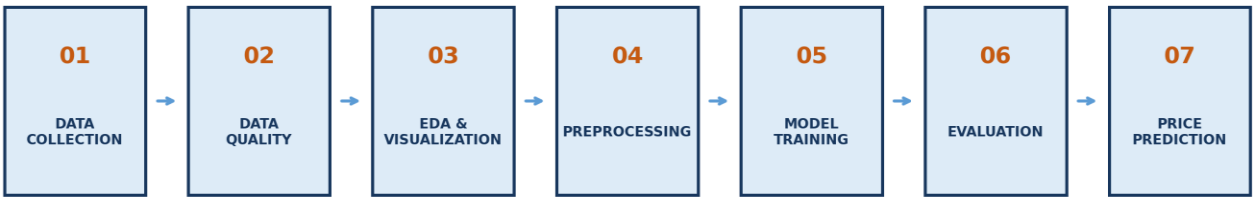
HOUSE PRICE PREDICTION USING MACHINE LEARNING

PROJECT REPORT

Student Name :	Pandhula Sinduja
Roll No :	160723733125
Branch :	Computer Science and Engineering
College :	Methodist College of Engineering and Technology
Project Domain :	Data Analytics + Artificial Intelligence

INTERNSHIP PROJECT • 2026

HOUSE PRICE PREDICTION — END-TO-END ANALYTICS WORKFLOW



Pandas • EDA • Preprocessing • Regression • Evaluation

ABSTRACT

House Price Prediction Using Machine Learning is a data-driven project designed to estimate house prices based on various property features. The price of a house depends on factors such as location, area, number of bedrooms, bathrooms, and property condition. This project uses historical housing datasets and machine learning algorithms to identify patterns between these features and property prices. The trained model predicts the estimated price of a house based on user-provided details. The model's performance is evaluated using appropriate metrics to measure prediction accuracy and error. The main objective is to provide a simple, efficient, and reliable house price estimation system that helps buyers, sellers, and real estate professionals make informed decisions.

1.INTRODUCTION

The real estate industry plays an important role in the economy, and estimating the value of a property is essential for both buyers and sellers. House prices vary according to several factors, including location, property size, number of rooms, construction quality, and nearby facilities. Traditional methods of price estimation often depend on manual analysis and expert judgment, which may take time and produce inconsistent results.

Machine learning provides an effective approach to analyzing housing data and predicting property prices. In this project, historical house price data is collected, cleaned, and processed to identify important features that influence house values. Machine learning algorithms, such as Linear Regression and Random Forest Regression, can be trained using this data to predict house prices for new properties. The performance of the trained model is evaluated using suitable metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared ((R^2)).The proposed system aims to make house price estimation faster, easier, and more consistent. It can support home buyers, property sellers, and real estate agents by providing estimated prices based on the available property information.

2.EXECUTIVE SUMMARY

This project presents an end-to-end data analytics and machine-learning solution for predicting residential house sale prices. The Ames Housing / Kaggle House Prices dataset is analyzed through data-quality checks, exploratory visualization, preprocessing, regression modeling and evaluation.The solution is designed as a reproducible internship project: the Jupyter Notebook contains the complete executable workflow, requirements.txt defines dependencies, and the README documents the project for GitHub.

3. Problem Statement

House prices vary depending on several factors, such as location, area, number of bedrooms, bathrooms, and available facilities. Predicting the correct price of a house is difficult using traditional methods because these factors influence property values differently. The main objective of this project is to develop a machine learning model that analyzes historical housing data and predicts the estimated price of a house based on its features. This system helps buyers, sellers, and real estate professionals make informed decisions by providing accurate and reliable house price estimates.

4. Objectives

- Understand a real-world housing dataset and its data quality.
- Explore relationships between property attributes and sale price.
- Handle missing numerical and categorical observations systematically.
- Transform categorical variables into machine-learning-ready features.
- Train and compare three regression algorithms.
- Evaluate predictions using MAE, RMSE, R^2 and RMSLE.
- Create visual outputs suitable for analytics interpretation and reporting.

5. Dataset Overview

The Kaggle House Prices: Advanced Regression Techniques dataset contains 1,460 training properties and 79 explanatory variables. The target variable is SalePrice.

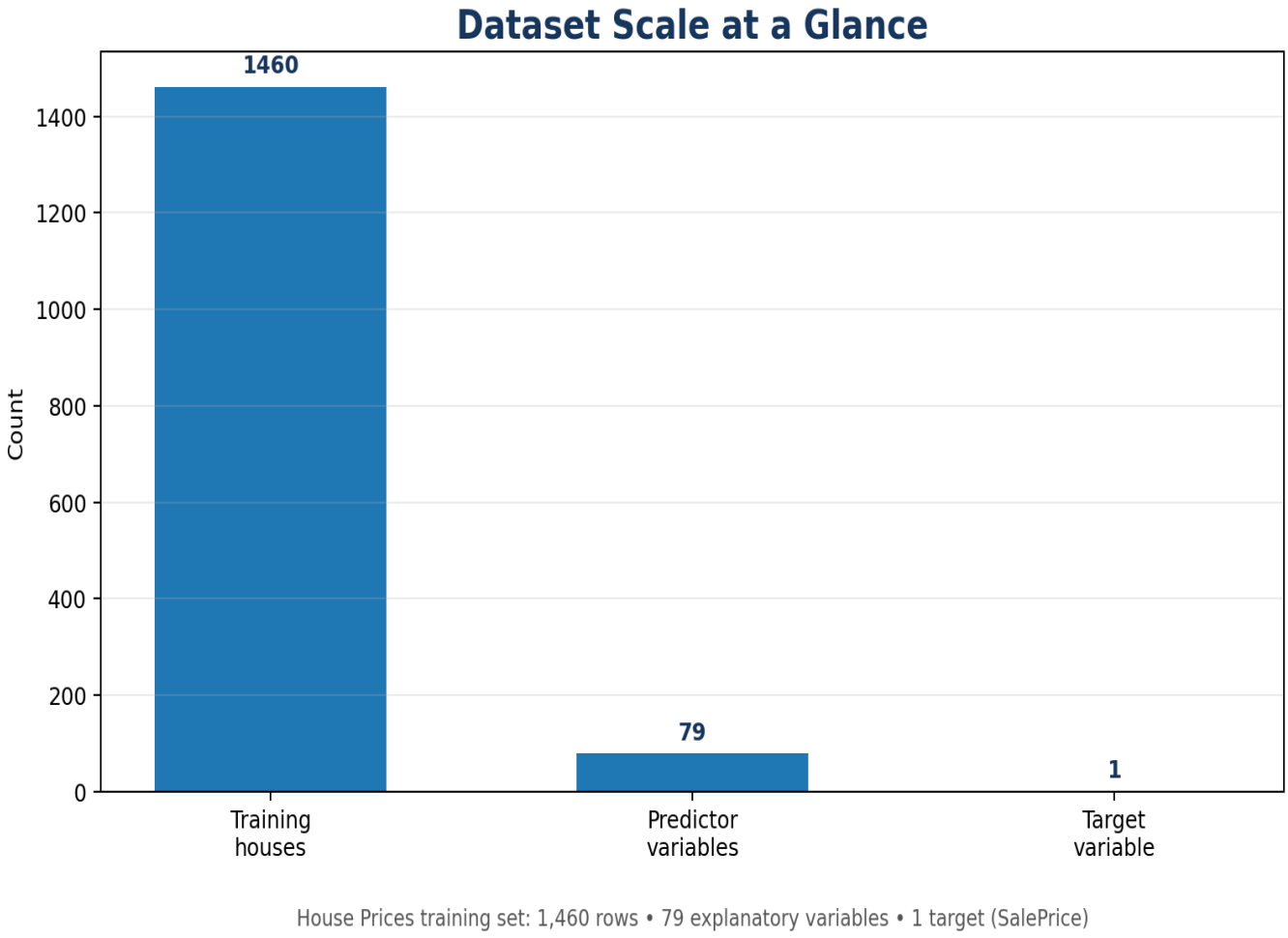


Figure 1. Dataset scale used by the project.

Dataset Item	Value	Purpose
Training properties	1,460	Model development
Explanatory variables	79	Property characteristics
Target variable	SalePrice	Regression prediction

6. Key Features

- OverallQual — overall material and finish quality
- GrLivArea — above-ground living area
- GarageCars — garage capacity
- TotalBsmtSF — total basement area
- 1stFlrSF — first-floor area
- YearBuilt — original construction year
- FullBath — number of full bathrooms
- TotRmsAbvGrd — rooms above grade

7. Project Workflow

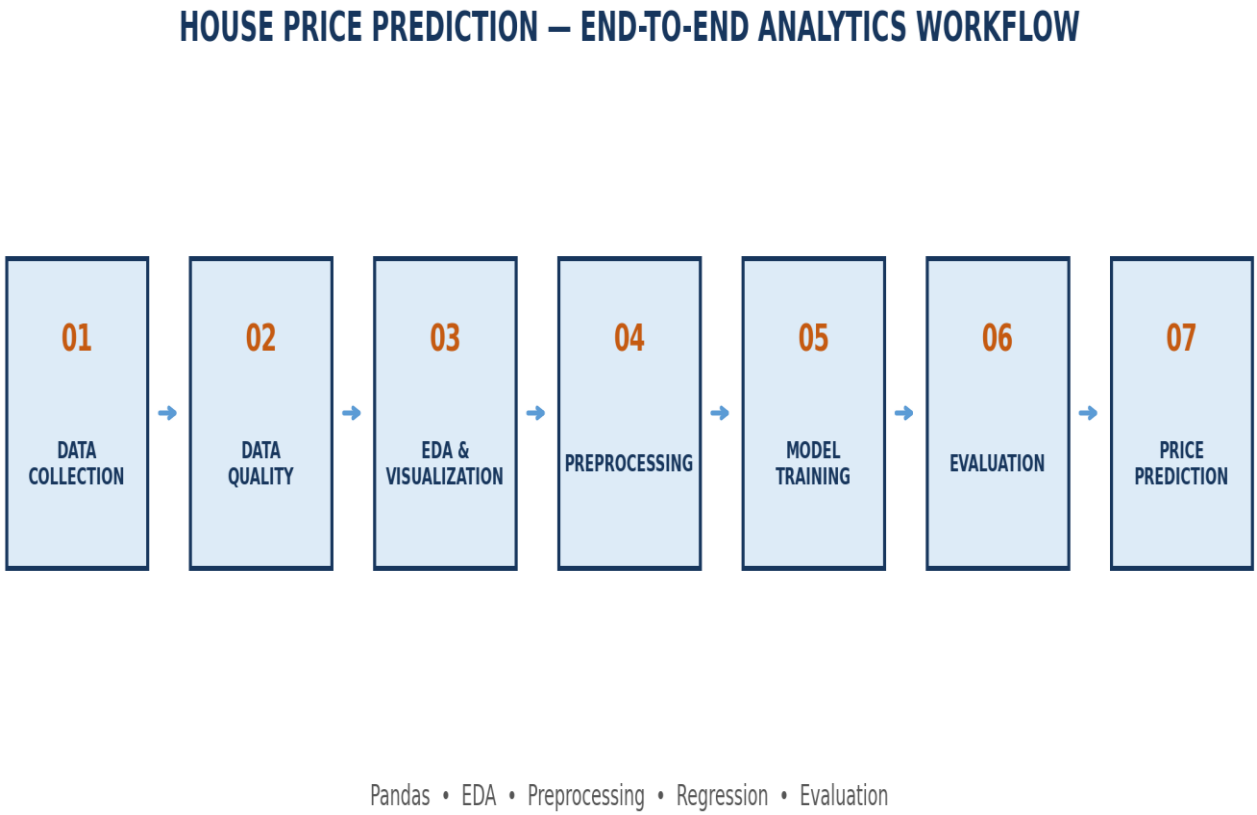


Figure 2. End-to-end analytics and machine-learning workflow.

8. System Architecture

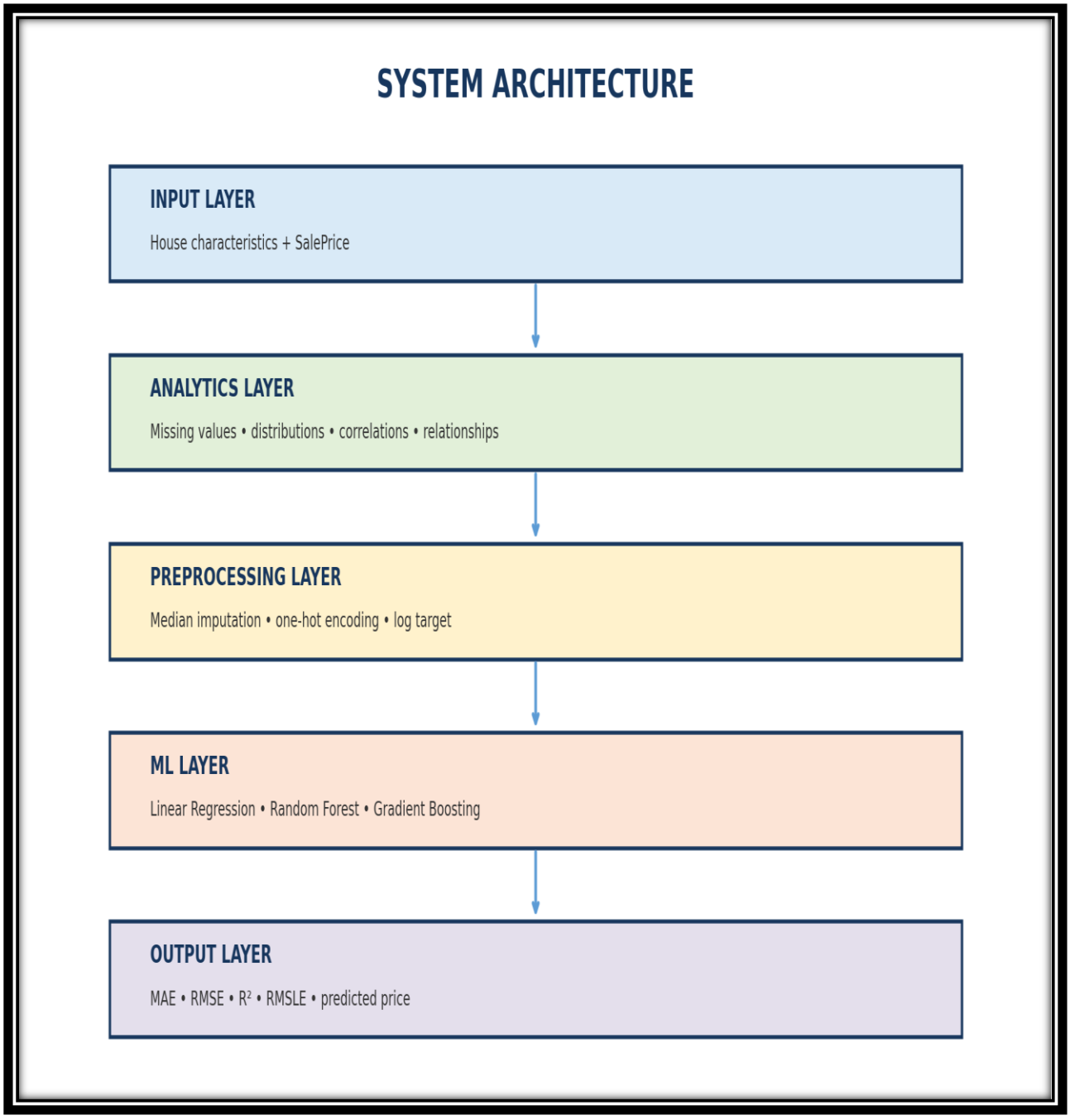


Figure 3. Layered system architecture.

9. Data Preprocessing

The notebook uses a Scikit-learn ColumnTransformer and Pipeline so that preprocessing is applied consistently to training and validation data.

- Numerical missing values → median imputation.
- Categorical missing values → most-frequent imputation.
- Categorical variables → One-Hot Encoding with unknown categories ignored.
- Identifier column Id → removed because it is an identifier rather than a meaningful predictive attribute.
- SalePrice → log1p transformation during training; predictions are converted back with expm1.

11. Exploratory Data Analysis

The executable notebook produces the actual dataset-based graphs during Run All. The main EDA views are SalePrice distribution, missing-value analysis, living-area versus price, and a correlation heatmap.

SalePrice Distribution Shows target distribution and skewness.	**Missing-Value Analysis** Highlights features requiring imputation.
GrLivArea vs SalePrice Explores the relationship between living area and price.	**Correlation Heatmap** Shows numerical relationships among important variables.

13. Machine Learning Models

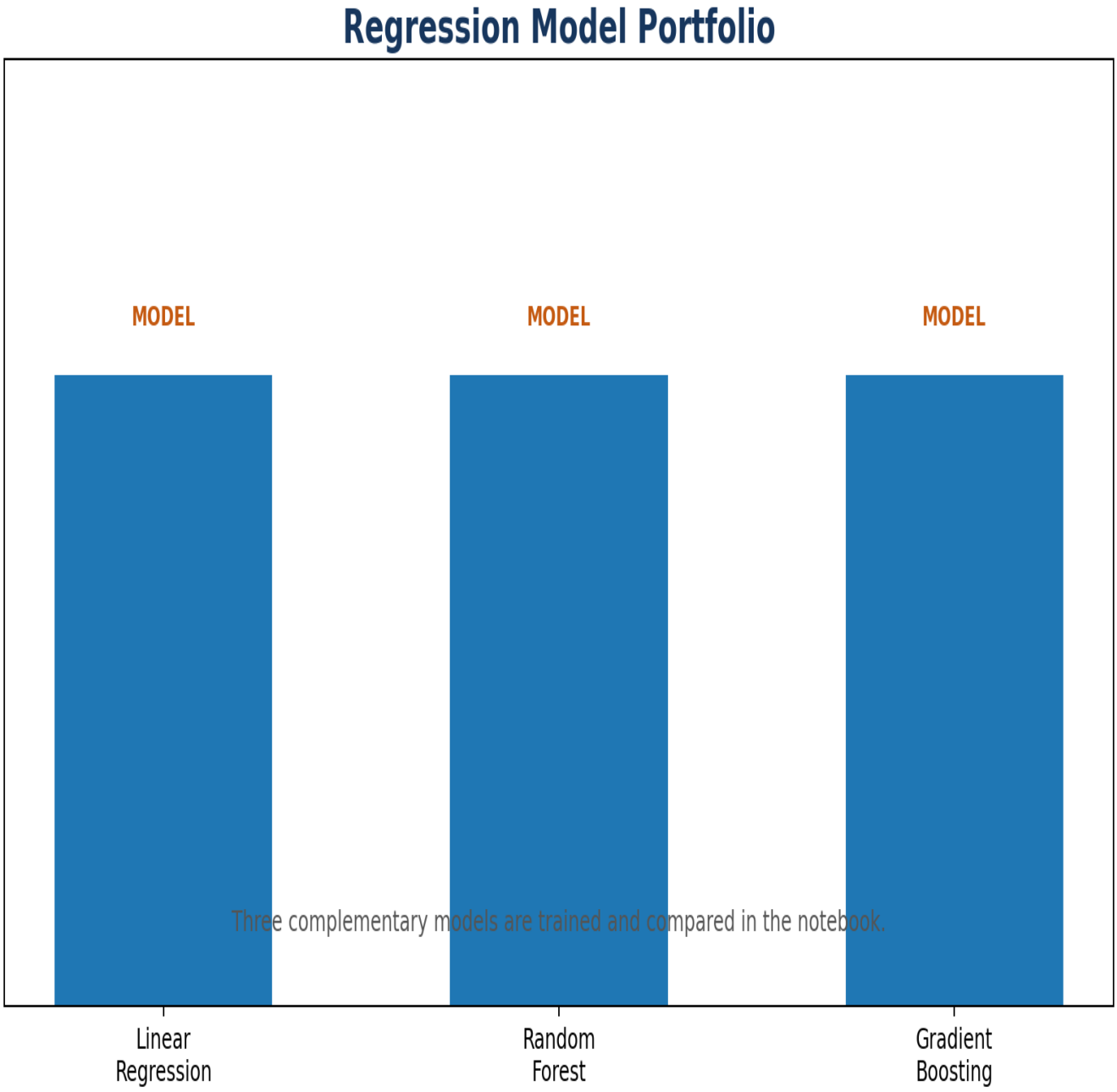


Figure 4. Three-model regression portfolio used for comparison.

- Linear Regression — interpretable baseline model.
- Random Forest Regressor — ensemble tree model for nonlinear relationships.
- Gradient Boosting Regressor — sequential tree ensemble designed to improve prediction error.

14. Evaluation Metrics

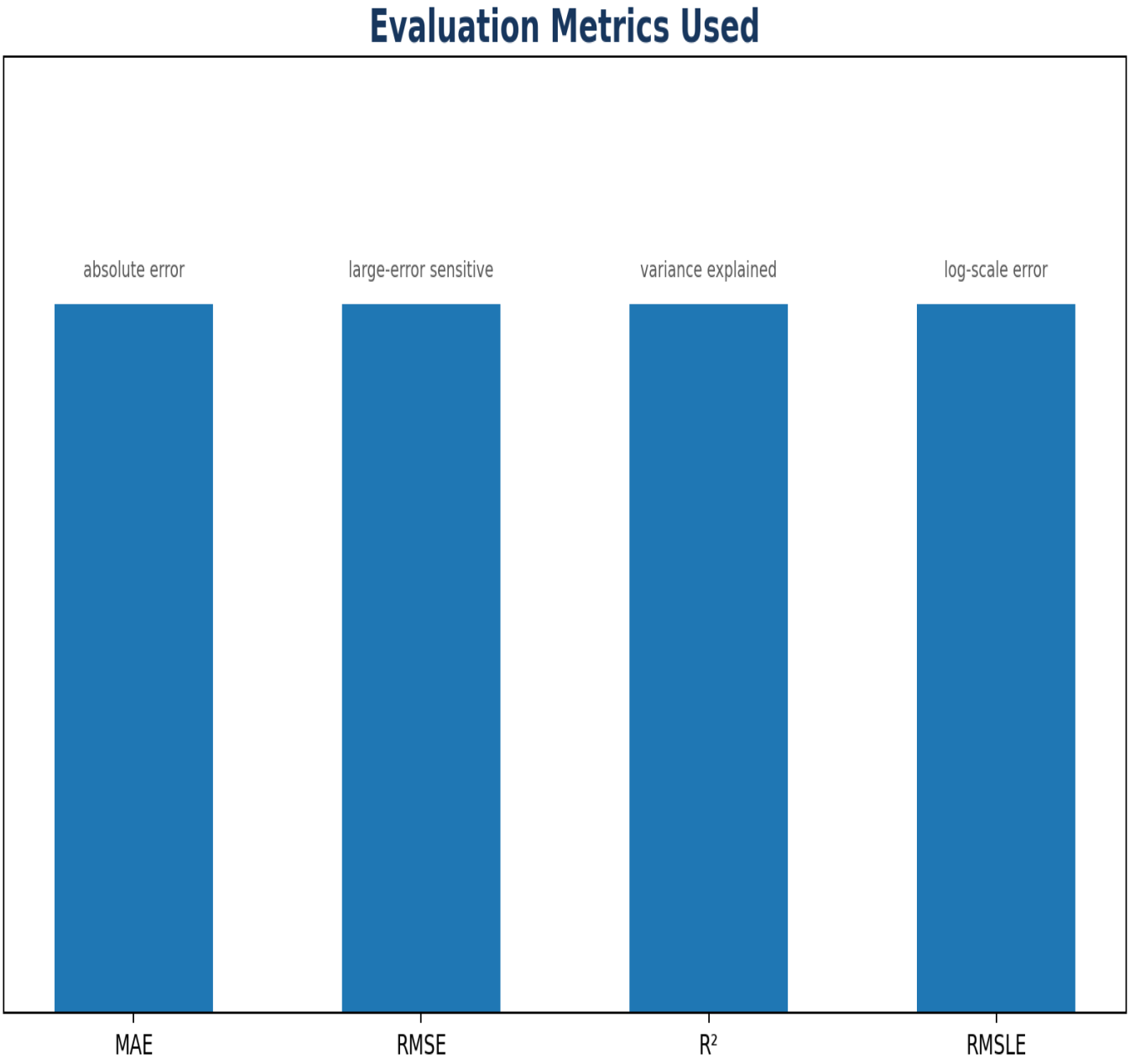


Figure 5. Evaluation metrics used by the notebook.

Metric	Meaning
MAE	Mean absolute prediction error.
RMSE	Penalizes larger errors more strongly.
R^2	Measures variance explained by the model.
RMSLE	Measures logarithmic prediction error.

15. Results and Output

The notebook generates the model comparison table and charts at execution time. Performance values are intentionally not hard-coded into this report, so the submitted results remain reproducible and match the actual execution environment.

Model	MAE	RMSE	R^2	RMSLE
Linear Regression	Generated on Run All	Generated on Run All	Generated on Run All	Generated on Run All
Random Forest	Generated on Run All	Generated on Run All	Generated on Run All	Generated on Run All
Gradient Boosting	Generated on Run All	Generated on Run All	Generated on Run All	Generated on Run All

16. Business Applications

- Preliminary property valuation and price estimation.
- Real-estate analytics and market research.
- Housing investment analysis.
- Property-feature and price relationship analysis.
- Data-driven decision support.

17. Limitations

- Historical data may not fully represent current market prices.
- Predictions depend on the quality and coverage of the training dataset.
- The model is an analytical estimate, not a professional property valuation.
- External market conditions are not explicitly modeled.

Future Enhancements

- Hyperparameter tuning with GridSearchCV or RandomizedSearchCV.
- Cross-validation for more robust evaluation.
- XGBoost, LightGBM and CatBoost comparison.
- SHAP-based explainability.
- Interactive Streamlit dashboard.
- Cloud deployment and model monitoring.

16. Conclusion

House Price Prediction project demonstrates a complete analytics-to-machine-learning workflow. It combines data understanding, cleaning, visualization, feature preparation, model comparison and evaluation into a single reproducible notebook suitable for the IBM SkillsBuild Data Analytics with AI internship.

References

- Kaggle — House Prices: Advanced Regression Techniques

https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data

- Scikit-learn documentation

https://scikit-learn.org/